

Total No. of Questions : 6]

SEAT No. :

**P207**

[Total No. of Pages : 2

**APR - 17/TE/Insem. - 43**

**T.E. (Computer Engineering)**

**DIGITAL SIGNAL PROCESSING APPLICATIONS**

**(2012 Pattern) (Semester - II) (310253)**

*Time : 1 Hour]*

*[Max. Marks : 30*

*Instructions to the candidates:*

- 1) *Attempt Q1 or Q2, Q3 or Q4, Q5 or Q6.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Assume suitable data if necessary.*

**Q1)** a) State the mathematical models used to represent a DT system. Define these models with mathematical form. **[5]**

b) Define the Impulse Response of a DT system and show that for a causal system

$$h(n) = 0 \text{ for } n < 0 \quad \mathbf{[5]}$$

OR

**Q2)** a) A CT signal having frequency 50 Hz is sampled at a rate of 1200 samples/sec. Obtain -

- i) Number of samples per cycle.
- ii) Digital/Discrete frequency  $f$  and  $\omega$ .
- iii) Minimum sampling rate to avoid aliasing effect.
- iv) Period of a DT signal. **[5]**

b) State the Linearity, causality and stability properties of a DT system. **[5]**

**Q3)** a) State and prove the time reversal property of Fourier Transform(FT). **[5]**

b) What do you understand by 'Indexing in Bit-Reversal' in FFT? Draw the basic butterfly structure for DIF FFT algorithm and hence obtain the computational complexity of N point DFT. **[5]**

OR

**P.T.O.**

**Q4)** a) Perform following circular shifting operations on a given DT signal  $x(n)=\{4,2,-1,3\}$  with  $N = 4$  and  $N = 5$ .

i)  $x((n-2))_N$

ii)  $x((n+1))_N$  [5]

b) What is the significance of 'N' in N point DFT? Define N point DFT by means of twiddle factor W and compute twiddle factors for  $N = 4$ . [5]

**Q5)** a) Define ROC of ZT. How many possible ROCs a single ZT may have? Give one example. [5]

b) Obtain ZT of a DT signal using ZT properties where,  
 $x(n) = n.u(n-1)$  Specify the ROC. [5]

OR

**Q6)** a) Draw a Pole Zero plot for a system described as-

$$y(n) = x(n) - x(n-1) + 0.2y(n-1) + 0.15y(n-2)$$
 [5]

b) Define the term system function  $H(Z)$ . Express it in the form of pole zero system and define it for FIR and IIR system. [5]

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